Math **163** Name (PLEASE PRINT) Capn Orangebeard

Exam **2C** **2020 Fall**/Spring/Summer I/Summer II Score _____

Please fill in the following table checking boxes corresponding to parts of problems that you have completed and submitted and an expected grade. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, as per the directions of the problem. Ideally you should know exactly what you have done correctly and what grade you should receive. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	4	1	1							6	
1b	3	3								6	
1c											
2a	6									6	
2b	6									6	
2c											
3a	1	1	4							6	
3b	1	1								2	
3c											
								Total		32	

Instructions:

Your J-number = 0099 0999

Keep every 0 in your J-number and replace every non-zero number in your J-number by 1 to get

Your (truth-value) **T-number**:= 0011 0111

Define $\forall k \in 1..8$,

T_k := the k[th] entry in your **truth-value J-number** counting from the left. Therefore,

$T_1 :=$ 0

$T_2 :=$ 0

$T_3 :=$ 1

$T_4 :=$ 1

$T_5 :=$ 0

$T_6 :=$ 1

$T_7 :=$ 1

$T_8 :=$ 1

Next construct an expression $G(w, x, y, z)$ as follows:

$\alpha(T_1, T_2, T_3, T_4)$

$= (T_1)(w) + (T_2)(x') + (T_3)(y) + (T_4)(z')$

$= (0)(w) + (0)(x') + (1)(y) + (1)(z')$

$= y + z'$

$\beta(T_3, T_4, T_5, T_6)$

$= \left(\left((T_3)w\right)'\right)\left(\left((T_4)x\right)'\right)\left(\left((T_5)y\right)'\right)\left(\left((T_6)z\right)'\right)'$

$= ((1)w)')(((1)x)')((0)y)')((1)z)')$

$= ((w')(x')(1)(z'))'$

$= w + x + z$

$\gamma(T_5, T_6, T_7, T_8)$

$= (T_5)(w') + (T_6)(x) + (T_7)(y') + (T_8)(z)$

$= 0(w') + 1(x) + 1(y') + 1(z)$

$= x + y' + z$

=

$G(w, x, y, z)$

$= \big(\alpha(T_1, T_2, T_3, T_4)\big)\big(\beta(T_3, T_4, T_5, T_6)\big)\big(\gamma(T_5, T_6, T_7, T_8)\big)$

$= (y + z')(w + x + z)(x + y' + z)$

=

6 (1a) Find $\varphi(p,q,r)$ in terms of: p,q and r if $TT\big(\varphi(p,q,r)\big)$ is given below. Find corresponding expressions $F(A,B,C)$ in ***Sets*** and $E(x,y,z)$ in ***BA*** on the facing page and record the answers on this page.

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	$T_1 =$ 0	0	0	0	0	0	0	0	0
0	0	1	$T_2 =$ 0	0	0	0	0	0	0	0	0
0	1	0	$T_3 =$ 1	0	0	0	0	1	0	0	0
0	1	1	$T_4 =$ 1	0	0	0	1	0	0	0	0
1	0	0	$T_5 =$ 0	0	0	0	0	0	0	0	0
1	0	1	$T_6 =$ 1	0	0	1	0	0	0	0	0
1	1	0	$T_7 =$ 1	0	1	0	0	0	0	0	0
1	1	1	$T_8 =$ 1	1	0	0	0	0	0	0	0

4(i) $\varphi(p,q,r)$

$= (p \wedge q \wedge r) \vee (p \wedge q \wedge (\neg r)) \vee (p \wedge (\neg q) \wedge r) \vee ((\neg p) \wedge q \wedge r) \vee ((\neg p) \wedge q \wedge (\neg r))$

1 (ii) $F(A,B,C)$

$= (A \cap B \cap C) \cup (A \cap B \cap (C^c)) \cup (A \cap (B^c) \cap C) \cup ((A^c) \cap B \cap C) \cup ((A^c) \cap B \cap (C^c))$

1 (iii) $E(x,y,z)$

$= xyz + xyz' + xy'z + x'yz + x'y'z'$

$$
\begin{aligned}
& \varphi(p,q,r) \\
= \; & \varphi_1 \vee \varphi_2 \vee \varphi_3 \vee \varphi_4 \vee \varphi_5 \\
= \; & (p \wedge q \wedge r) \\
& \vee (p \wedge q \wedge (\neg r)) \\
& \vee (p \wedge (\neg q) \wedge r) \\
& \vee ((\neg p) \wedge q \wedge r) \\
& \vee ((\neg p) \wedge q \wedge (\neg r))
\end{aligned}
$$

$$
\begin{aligned}
& F(A,B,C) \\
= \; & (A \cap B \wedge C) \cup (A \cap B \cap (C^c)) \cup (A \cap (B^c) \wedge C) \cup ((A^c) \cap B \cap C) \cup ((A^c) \cap B \cap (C^c))
\end{aligned}
$$

$$
\begin{aligned}
& E(x,y,z) \\
= \; & xyz + xyz' + xy'z + x'yz + x'yz'
\end{aligned}
$$

---

6 (1b) Use the consensus method to find the following for your answer from (1a):

$$E(x,y,z) = xyz + xyz' + xy'z + x'yz + x'yz'$$

3 (i) the prime implicants of E: $PI\ (E) = \{y, xz\}$

3 (i) the minimal sum-of-products form: $MSOP\ (E) = xz + y$

1b. i.0

E

$= \cancel{xyz} + \cancel{xyz'} + xy'z + x'yz + x'y'z'$ $\quad Cns(xyz, xyz') = xy$

$= \cancel{xyz} + \cancel{xyz'} + xy'z + x'yz + x'y'z' + \cancel{xy}$ $\quad (xy \subseteq xyz,\ xy \subseteq xyz')$

$= xy'z + \cancel{x'yz} + \cancel{x'yz'} + xy$ $\quad Cns(x'yz, x'yz') = x'y$

$= xy'z + \cancel{x'yz} + \cancel{x'yz'} + xy + \cancel{x'y}$ $\quad (x'y \subseteq x'yz,\ x'y \subseteq x'yz')$

$= xy'z + \cancel{xy} + \cancel{x'y}$ $\quad Cns(xy, x'y) = y$

$= xy'z + \cancel{xy} + \cancel{x'y} + \cancel{y}$ $\quad (y \subseteq xy,\ y \subseteq x'y)$

$= \cancel{xy'z} + \cancel{y}$ $\quad Cns(xy'z, y) = xz$

$= \cancel{xy'z} + y + \cancel{xz}$ $\quad (xz \subseteq xy'z)$

$= xz + y$

i.1

$$\frac{(xz \not\subseteq y) \quad (y \not\subseteq xz) \quad \cancel{Cns}(xz, y)}{PI(E) = \{xz, y\}}$$

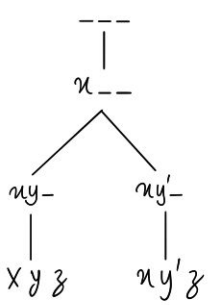
ii.0

E

$= xz \mid + y$ $\quad = x(1)z \mid + (1)y(1)$ $\quad = x(y+y')z \mid + (x+x')y(z+z')$ $\quad = xyz(11) + xyz' + x'yz + x'yz' \mid + xyz(12) + xyz' + x'yz + x'yz' \mid$ $\quad = xz \mid + y \mid$ $\quad = MSOP(E)$

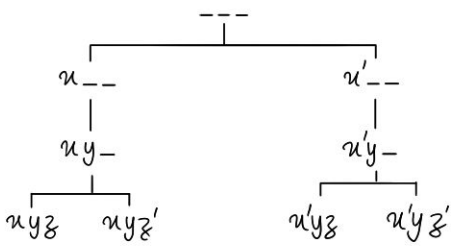
ii.1

Tree



ii.2

Tree



6 (2a) For the expression from page 53 below:

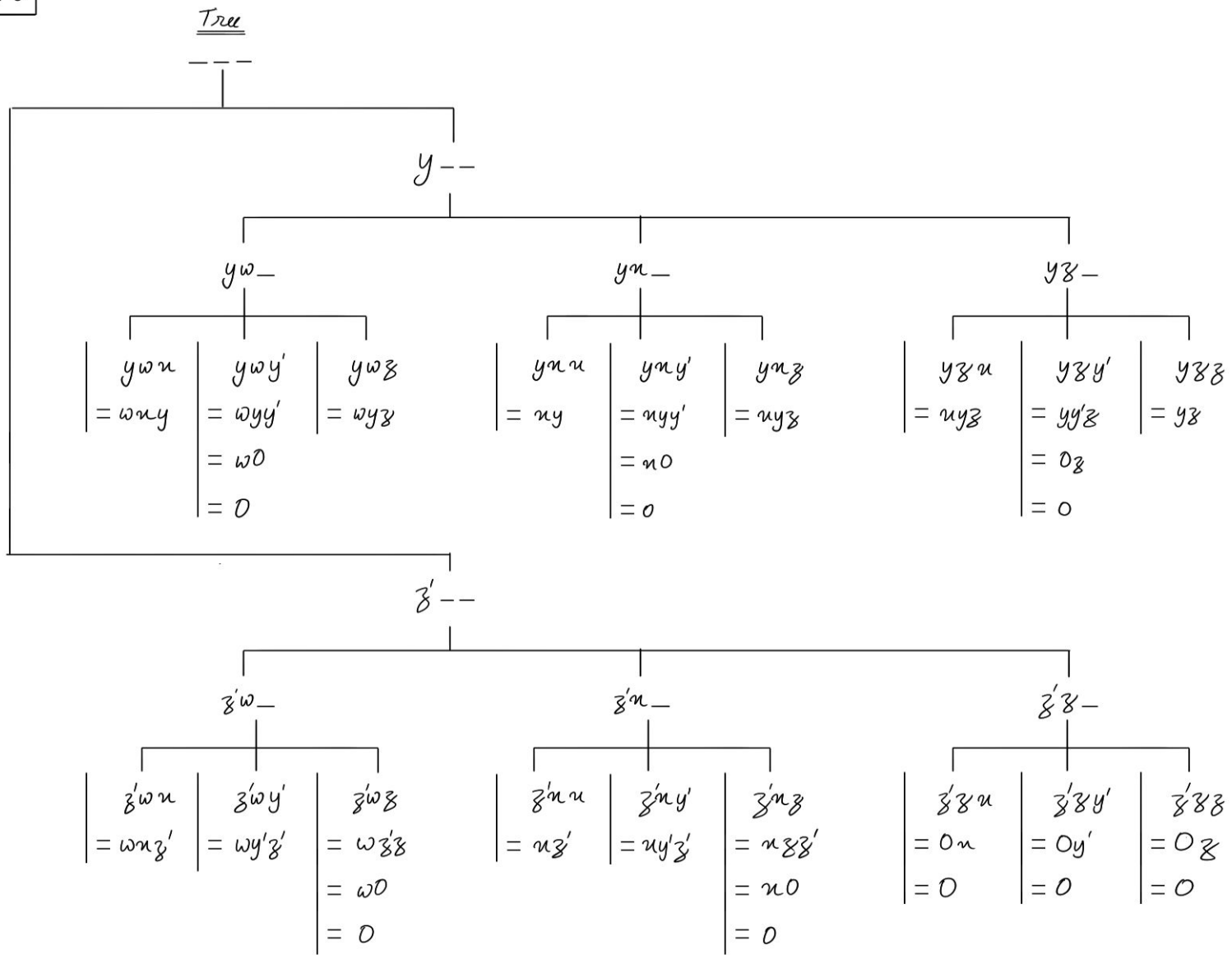
$$G(w,x,y,z) = (y+z')(w+x+z)(x+y'+z)$$

find a sum-of-products form, $SOP(G)$

$$SOP(G) = wy'z' + xy + xz' + yz$$

Show the tree on the facing side.

2a. 0



1

$$Cz(w,x,y,z)$$

$$= wxy + 0 + wyz + xy + 0 + xyz + xyz + 0 + yz + wxz' + wy'z' + 0 + xz' + xy'z' + 0 + 0 + 0 + 0$$

$$= \underline{wxy} + \underline{wyz} + \underline{xy} + \underline{xyz} + \underline{xyz} + \underline{yz} + \underline{wxz'} + wy'z' + \underline{xz'} + \underline{xy'z'}$$

$$= wy'z' + xy + xz' + yz$$

$(xy \subseteq wxy,\ xy \subseteq xyz,\ yz \subseteq wyz,\ xz' \subseteq wxz',\ xz' \subseteq xy'z')$

2

$wy'z' \not\leq xy$ $wy'z' \not\leq yz$ $wy'z' \not\leq xz'$

$xz' \not\leq wy'z'$ $xz' \not\leq xy$ $xz' \not\leq yz$

$yz \not\leq wy'z'$ $yz \not\leq xy$ $yz \not\leq xz'$

$xy \not\leq wy'z'$ $xy \not\leq yz$ $xy \not\leq xz'$

3

$$Cz(w,x,y,z)$$

$$= wy'z' + xy + xz' + yz$$

$$= SOP(Cz)$$

6110

6 (2b) For the expression below (**which is the same as in (2a)**):

$$G(w,x,y,z) = (y+z')(w+x+z)(x+y'+z)$$

find the complete sum-of-products form, $CSOP(G)$

$$\begin{aligned} CSOP(G) = \; & wxyz \\ & + wxyz' \\ & + wxy'z' \\ & + wx'yz \\ & + wx'y'z' \\ & + w'xyz \\ & + w'xyz' \\ & + w'xy'z' \\ & + w'x'yz \end{aligned}$$

26.0

SOP(G)

$= wy'z' = w(1)y'z' = w(x+x')y'z' = wxy'z'(11) + wx'y'z'$

$+ xy \quad + (1)xy(1) \quad + (w+w')xy(z+z') \quad + wxyz(21) + wxyz'(31) + w'xyz(41) + w'xyz'(51)$

$+ yz \quad + (1)(1)yz \quad + (w+w')(x+x')yz \quad + wxyz(22) + wx'yz + w'xyz(42) + w'x'yz$

$+ xz' \quad + (1)x(1)z' \quad + (w+w')x(y+y')z' \quad + wxyz'(32) + wxy'z'(12) + w'xyz'(52) + w'xy'z'$

$= wxy'z' + wx'y'z' = wxyz = CSOP(G)$

$+ wxyz + wxyz' + w'xyz + w'xyz'$

$+ w'xy'z' + wx'yz + w'x'yz$

$+ wxyz'$

$+ wxy'z'$

$+ wx'yz$

$+ wx'y'z'$

$+ w'xyz$

$+ w'xyz'$

$+ w'x'yz'$

$+ w'x'y'z$

1

Tm

$yz\,--$

$yzw\,-$ | $yzw'\,-$

$yzwx$ = $wxyz$ | $yzwx'$ = $wx'yz$ | $yzw'x$ = $w'xyz$ | $yzw'x'$ = $w'x'yz$

2

Tm

$xz'\,--$

$xz'w\,-$ | $xz'w'\,-$

$xz'wy$ = $wxyz'$ | $xz'wy'$ = $wxy'z'$ | $xz'w'y$ = $w'xyz'$ | $xz'w'y'$ = $w'xy'z'$

3

Tm

$xy\,--$

$xyw\,-$ | $xyw'\,-$

$xywz$ = $wxyz$ | $xywz'$ = $wxyz'$ | $xyw'z$ = $w'xyz$ | $xyw'z'$ = $w'xyz'$

4

Tm

$wy'z'\,-$

$wy'z'x$ = $wxy'z'$ | $wy'z'x'$ = $wx'y'z'$

6130

6 (3a) We define:

$A := SOP(G(w,x,y,z)) =$ $wy'z' + xy + xz' + yz$

$B = MSOP(G(w,x,y,z)) =$ $wy'z' + xz' + yz$

Copy your answers from (1a)

Compute:

1 (i) $A_S = 4$ $A_L = 9$ $B_S = 3$ $B_L = 7$

1 (ii) On the basis of your answers to (i), circle the correct choice below.

$Smplr\ (A,B)$ $Smplr(B,A)$ Neither

Why?

4 (iii) Write down the version of $G(w,x,y,z)$ in PC below and find its **formation-tree on the facing side**.

Version of $G(w,x,y,z)$in PC

$= (r \vee (\neg s)) \wedge (p \vee q \vee s) \wedge (q \vee (\neg r) \vee s)$

3a.i.0

$$
\begin{aligned}
&A_s \\
&= \nu(\mathrm{Smnd}(\mathcal{A})) \\
&= \nu(\{wy'z', xy, xz', yz\}) \\
&= 4
\end{aligned}
$$

i.1

$$
\begin{aligned}
&A_\ell \\
&= \nu(M_{\mathrm{slfr}}(A)) \\
&= \nu(\{w:1, x:2, y:2, y':1, z:1, z':2\}) \\
&= 9
\end{aligned}
$$

i.2

$$
\begin{aligned}
&B_s \\
&= \nu(\mathrm{Smnd}(B)) \\
&= \nu(\{wy'z', xz', yz\}) \\
&= 3
\end{aligned}
$$

i.3

$$
\begin{aligned}
&B_\ell \\
&= \nu(M_{\mathrm{slfr}}(B)) \\
&= \nu(\{w:1, x:1, y:1, y':1, z:1, z':2\}) \\
&= 7
\end{aligned}
$$

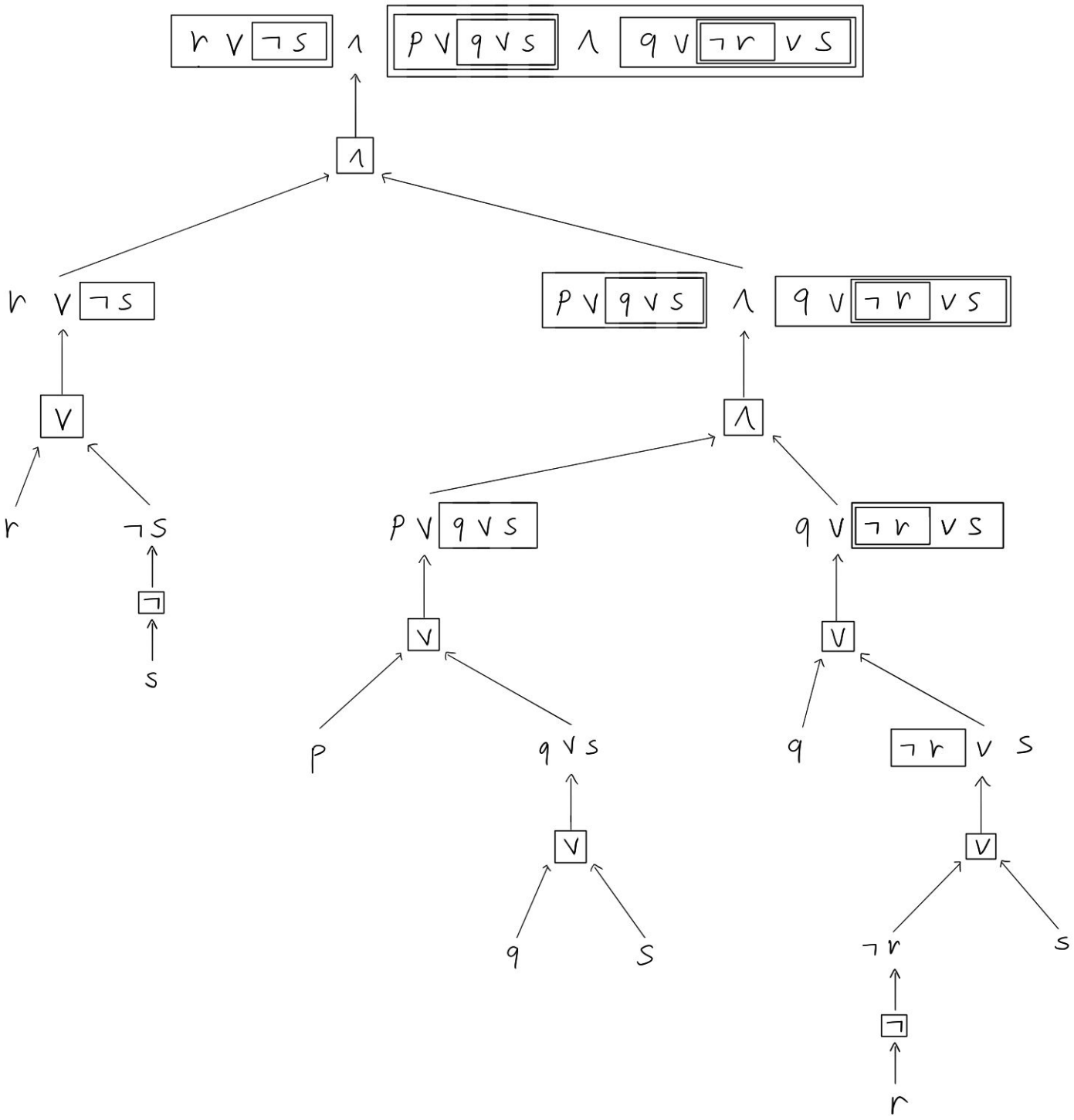
ii

$$
\frac{\big((B_\ell = 7 < 9 = A_\ell) \wedge (B_s = 3 \leq 9 = A_\ell)\big)}{\cancel{\mathrm{Smplr}}(A,B) \quad \wedge \quad \mathrm{Smplr}(B,A)}
$$

iii.0

$$
\frac{G_2(w,x,y,z) = (y+z')(w+x+z)(x+y'+z)}{\begin{aligned}&\psi(p,q,r,s) \\ &= (r \vee (\neg s)) \wedge (p \vee q \vee s) \wedge (q \vee (\neg r) \vee s)\end{aligned}}
$$

3a. iii. 1



2 (3b) Find the versions: $H(A,B,C,D)$ of $G(w,x,y,z)$

and $K(A,B,C,D)$ of $MSOP\big(G(w,x,y,z)\big)$

in Sets.

1 (i) $H(A,B,C,D)$

$= \quad (C \cup (D)^c) \cap (A \cup B \cup D) \cap (B \cup (C)^c \cup D)$

1 (i) $K(A,B,C,D)$

$= \quad (A \cap (C)^c \cap (D)^c) \cup (B \cap (D)^c) \cup (C \cap D)$

3b.i

$$G(w,x,y,z) = (y+z')(w+x+z)(x+y'+z)$$

$$H(A,B,C,D) = (C \cup (D)^c) \cap (A \cup B \cup D) \cap (B \cup (C)^c \cup D)$$

ii

$$MSOP(G(w,x,y,z)) = wy'z' + xz' + yz$$

$$K(A,B,C,D) = (A \cap (C)^c \cap (D)^c) \cup (B \cap (D)^c) \cup (C \cap D)$$